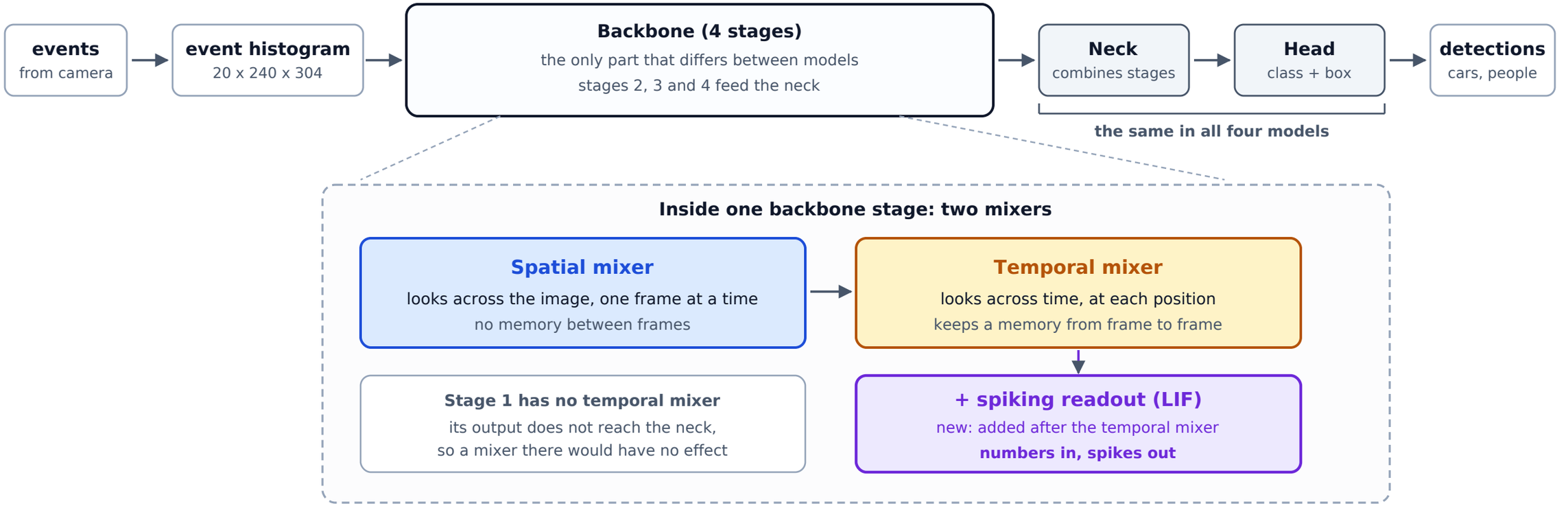
a The detector, shared by all models

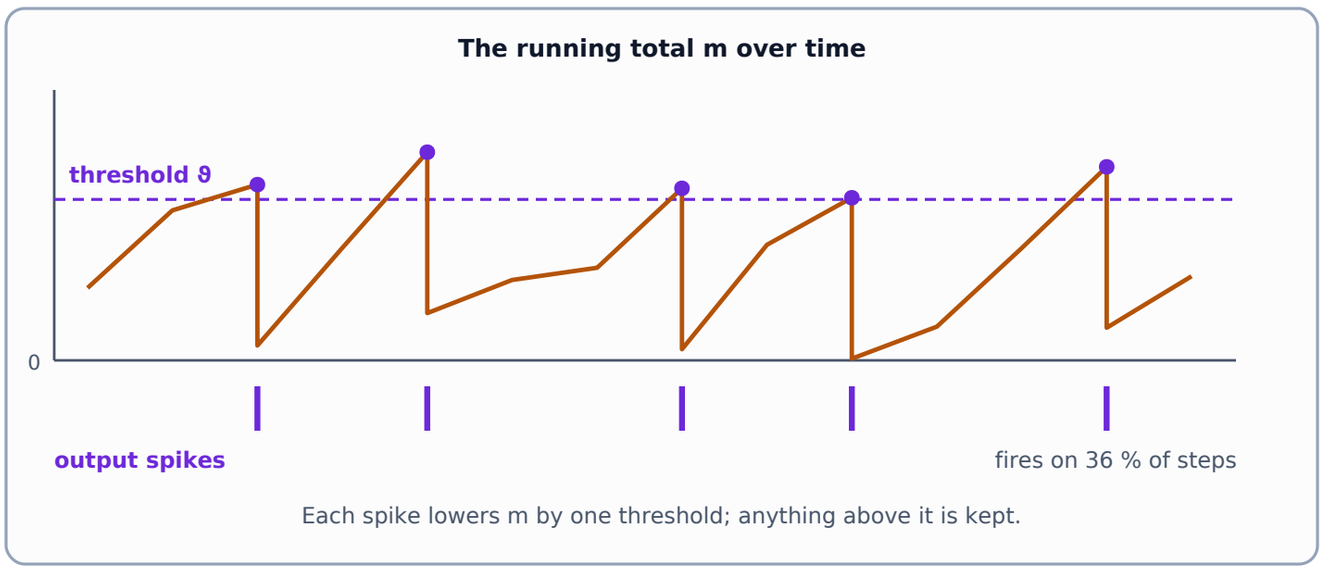


b How the spiking readout works (trace from the real code)

at every time step, for each value y from the mixer

m ← β m + y
add the input; the old total slowly leaks away

s = 1 if m ≥ ϑ, else 0
fire a spike when m reaches the threshold ϑ

m ← m − s ϑ
after a spike, take the threshold off m

Training uses a smooth stand-in for the step,
because the step itself gives no slope to learn from.